LABORATORY TITLE: USB IMAGE ACQUISITION AND HASH VERIFICATION

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STUDENT ID: 2025/FWSD/11435

COURSE NAME: COMPUTER AND DIGITAL FORENSICS

COURSE CODE: SBT-DF202 (LAB 2)

INSTRUCTOR NAME: MR. AMINU IDRIS

DATE OF SUBMISSION: AUGUST 29TH, 2026.

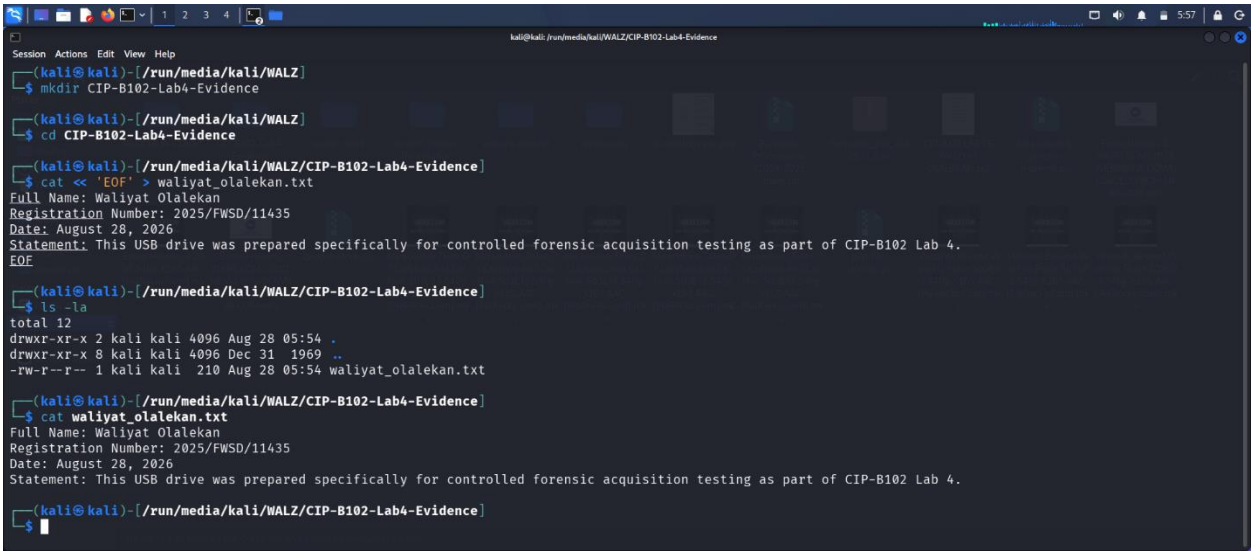
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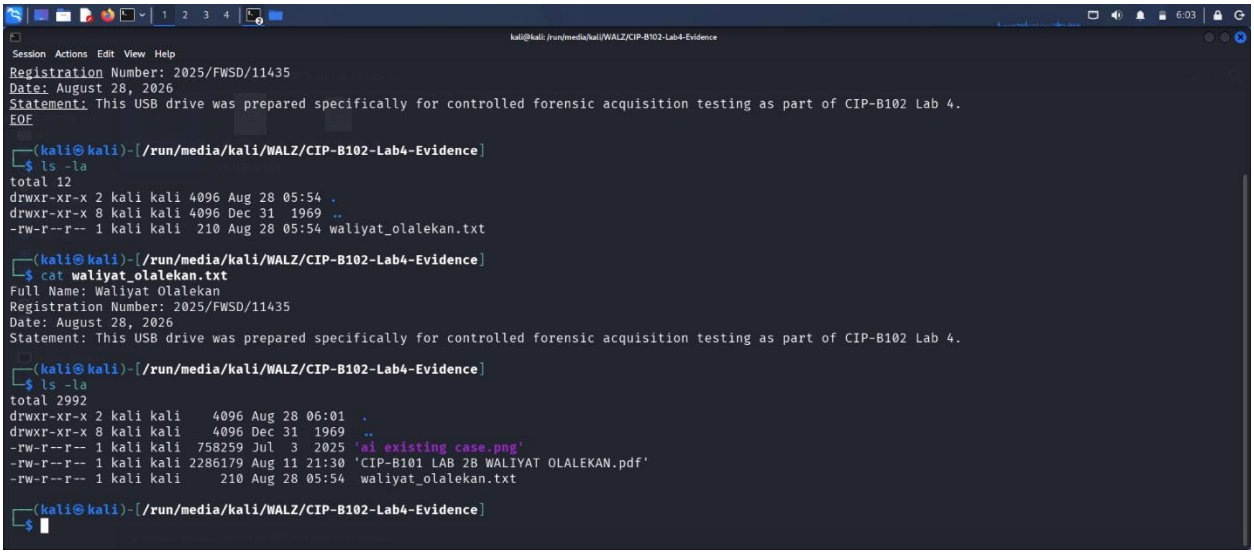
Lab Objective

The primary objective of CIP-B102 Lab 4 is to establish core competencies in low-level bit-stream disk acquisition, cryptographic integrity verification, non-destructive file system image validation, and forensic evidence logging. Specific learning outcomes include:

1. Target Device Isolation: Identifying target block storage devices and safely unmounting active partitions to prevent operating system write-contamination.
2. Physical Bit-Stream Acquisition: Generating an uncompressed raw physical image (.dd) of a physical USB drive using GNU dd.
3. Cryptographic Baseline Hashing: Calculating pre- and post-acquisition MD5 and SHA-256 cryptographic hashes digests directly against physical device nodes and acquired raw image files to verify 100% data integrity.
4. Non-Mount Image Validation: Utilizing Sleuth Kit command-line utilities (mmls, fls, icat) to inspect partition tables and parse internal metadata structures without kernel mounting.
5. Technical Risk Analysis: Evaluating operational hazards associated with low-level command-line tools (dd/dcfldd) and establishing best practices for digital evidence handling.

Part A - Prepare the USB Evidence





Part B - Complete Evidence Documentation

Case/Lab ID - CIP-B102-Lab4

Student name – Waliyat Olalekan

USB brand/description - SanDisk

USB capacity – 8 GB

Date and time prepared – 28th August 1:20pm

Acquisition method - Physical Bit-Stream Image (dd)

Tool and version - dd (Linux / Kali)

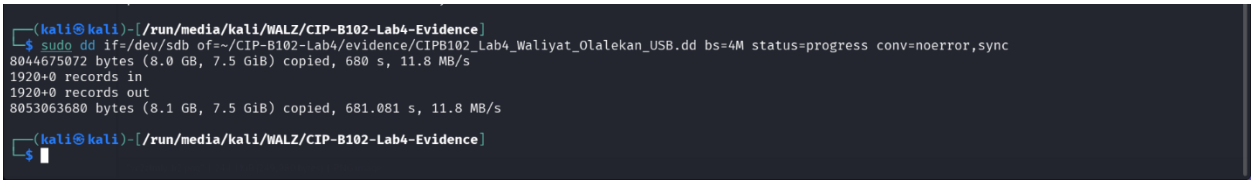
Image filename - CIPB102_Lab4_Waliyat_Olalekan_USB.dd

Image storage location - /home/kali/CIP-B102-Lab4/evidence/

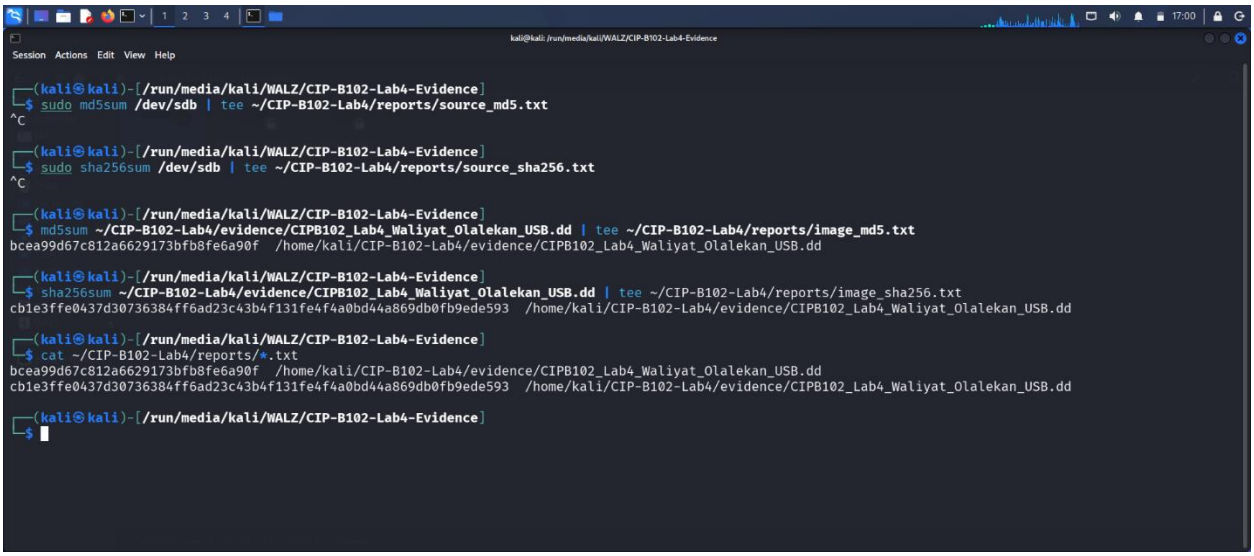
Hash algorithm used - MD5 and SHA-256

Part C - Acquire the USB Image

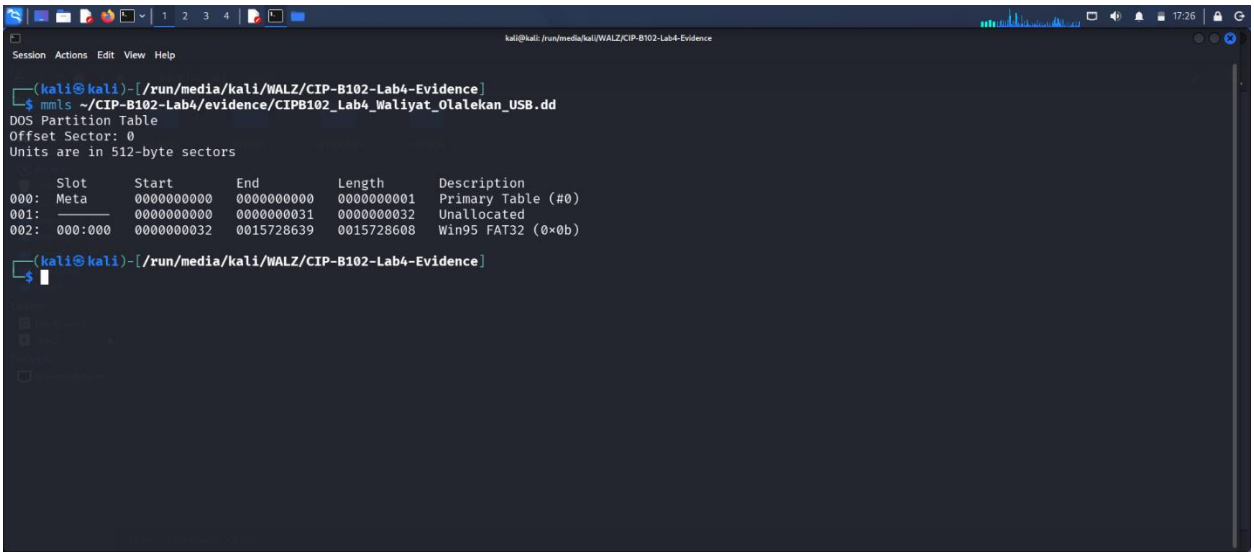
Option B - Linux/Kali using DD



Part D - Hash Verification



Part E - Validate the Acquired Image





Conclusion

The completion of this lab practice demonstrates practical competence in executing end-to-end physical bit-stream disk acquisitions. Key technical conclusions include:

1. Non-Destructive Acquisition: Unmounting active partitions before running dd prevents kernel write-contamination during block extraction.
2. Mathematical Authenticity: Matching pre- and post-acquisition SHA-256 and MD5 hashes confirms bit-stream parity between physical media and the .dd image file.
3. Advanced Artifact Recovery: Leveraging The Sleuth Kit (mmls, fls, icat) allows investigators to validate internal file artifacts within unmounted forensic images while bypassing host operating system mount dependencies.